

BHABHA UNIVERSITY, BHOPAL



Ph.D. Course Work (I) Syllabus

NH-12 Hoshangabad Road, Jatkhedi, Bhopal (M.P.)



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSE WORK

SUBJECT: PSYCHOLOGY

Research Methodology, Quantitative Techniques and Computer Applications

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHPC-1	PSYCHOLOGY	4	3	1	70	20	10

Unit- I: The meaning of methodology, The research process. Nature and types of variables. Techniques for controlling extraneous variables. Sources of bias. Types of Research : Descriptive, Relational, Experimental and Ex post facto. Methods of data collection : Questionnaire, interview, Observation, Case study.

Unit- II : Conceptual framework of experimental design : between groups and within group designs. Quantitative methods : Nonparametric and parametric : non parametric : Sign test, Mann Whitney U test and Chi Square. Parametric test : Test, One Way and Two Way Analysis of variance Newman Keuls, Duncan multiple range test.

Unit- III : Simple correlation, Biserial correction, Tetrachoric correlation, and partial correlation. Overview of simple and multiple regressions. Introduction to factor analysis.

Unit- IV : Defining Qualitative research. How is qualitative research conducted ? Issues in conducting of qualitative research, Non-reactive research or unobtrusive measures : The logic, purpose and definition and types of non-reactive research. Research report writing in APA style.

Unit-V : Introduction to Computer : History, Characteristics of Computer. Knowledge about internet use and search engines. Introduction to operating system-Windows. MS Office : MS Word, Excel, Power Point. Introduction to Statistical Package for Social Sciences (SPSS).

Recommended Readings:

Broota, K.D. (2003), Experimental design in Behavioral research. New Age International, New Delhi.

Kerlinger, F.N. (2000), Foundations of behavioral research. Wordsworth publication House : New

York. Swain, A.K. (2007), A text book on research methodology. Kalyani Publishers : New Delhi.

Newman, W.L. (1991), Social research methods: Quantitative and qualitative approaches. Allen Bacon : London.



BHABHA UNIVERSITY, BHOPAL

COURSE WORK FOR PH.D. STUDENT

SUBJECT: CHEMISTRY

PAPER-I

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHCH-1	CHEMISTRY	4	3	1	70	20	10

- 1. Fundamental Laboratory Techniques:** Basic Principles, Health and safety, working with liquids, Basic laboratory procedures-I, Basic laboratory procedures-II, Principles of solution chemistry, pH and buffer solutions.
- 2. The investigative approach:** Making and recording measurements, SI units and their use, Scientific method and design of experiments, Project work.
- 3. Analysis and Presentation data:** Using graphs, presenting data in tables, Hints for solving numerical problems, Descriptive statistics, choosing and using statistical tests, drawing chemical structures, chemometrics, computational chemistry.
- 4. Techniques of Quantitative and qualitative analysis of organic and inorganic compounds:** Theoretical bases, Acidometry Alkalimetry, Iodometry, redox titration, precipitation & complexometry, gravimetry, miscellaneous methods.
- 5. Spectrometry:** UV-Vis, IR, Mass, NMR, AAS, AES, GC-MS, HPLC- MS, Spectrofluorometry.
- 6. Thermal Methods of Analysis:** Principle and applications of TGA, DTA and DSC; Cyclic voltammeter.

Reference :-

1. Practical Skills in Chemistry, J.R. Dean, A.M. Jones, D. Holmes, R. Reed J. Weyers and A Jones, Pearson Education Ltd. (Prentice Hall) 2002.
2. OSU safety Manual 1.01.
3. Research Methodology, Methods and Techniques : C. R. Kothari,
4. Test, Measurements and Research, Methods in Behavioural Sciences : A.K. Singh.
5. Modern Methods of organic synthesis, William Carruthers and Iain Coldham, Cambridge University Press, Fourth Ed. 2004.
6. Experiments and Techniques in organic chemistry, D. Casto, C Johnson and M. Miller, Prentice-Hall.
7. Practical Pharmaceutical Chemistry, CBS Publishers ND Fourth Ed., In Part I & II by A.H. Beckett and J.B. Stenlake.
8. Elementary Practical organic chemistry Part I, II, III Arthur I. Vogel. (2nd edition) CBS Publishers & Distributors.



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSE WORK-I

SUBJECT: MICROBIOLOGY

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHMB-1	MICROBIOLOGY	4	3	1	70	20	10

Biosafety Lab (BSL) conditions: Rules for maintaining the Biosafety in the Microbiology laboratory, Making the aseptic, sterilized and dust free conditions in the lab for prevention of unwanted contamination of the sterilized lab from outer Sources.

Culture, Characterization and Identification: Awareness about of types of Culture media for both aerobic and anaerobic microbes under extreme conditions. Knowledge of Biochemical, Serological and Molecular biological aspects of techniques are essential

Laboratory Animal Care: Maintenance of Germ free animals (rat, mice, guinea pigs) and , inoculation via various route, immunization schedule and their knowledge , handling of WISTAR rats, Swiss albino, Balb/C mice and Rabbits is required, Also knowledge of Cell and Animal tissue culture.

Research in Biology:- Microbiological, Serological and Molecular biology related problems and assumptions, Search of Antimicrobial herbal products, Immunomodulation, Secondary metabolical product of Microbes study problems, Reference and literature search, Records and presentation of data, Biological literature, Technical papers, Abstracts, Reprints and other literature

Principles And Application: PCR and RT-PCR, Gel documentation, RAPD and RFLP, PAGE and Western and Southern blotting, Special staining, and Microscopy, Spectrophotometer, Ultra violet and infra red Spectrophotometer, ELISA, Geldoc, PAGE, Thermocycler, Microscope, Paper electrophoresis and Gel electrophoresis.

Centrifugation techniques: Principle, Types of centrifuges and their applications

General Principles and techniques of Histology: Microtomy and Histopathology, Cryotomy and Microphotography.

Quantitative Estimations: Principle, methods and applications of Protein, Lipids, Enzymes, Free Radicals, Antigen and Antibody estimation

Biostatistics: Mean, Median, Mode, Histograms, Frequency curve, Frequency Polygons, Bar diagrams, Pie-diagrams Standard Deviation and Standard Error Normal Distribution and Binomial Distribution.

Test of Significance Based on large Samples and Small Samples: t-Test, Chi-square test, ANOVA, Basics of correlation and Regression analysis.

Computer applications: Basic idea of computers: MS Word, power point and Excel,

Bioinformatics: Definition, role, scope and limitations.

Biological data and databases: Biological data type, classification of biological Data base, Sequence databases: Gen Bank, SWISS-PROT, Specialized databases: KEGG, ENZYME Secondary nucleotide and protein sequence databases: CUTG, PROSITE

BOOKS FOR REFERENCE:

Cell and Molecular Biology- De Robertis and De Robertis

Textbook of Medical Physiology- Guyton and Hall

Cell and Molecular Biology- P.K.Gupta

Fundamentals of Statistics- D.M. Elhance

Biostatistics and Computer Application- Dr. R. Goswami

Immunology - Evan M Roit

Microbiology- Peljar

Manual of Microbiology- Cruickshank, Monica Chelmsburgh.



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSEWORK -I SUBJECT- MANAGEMENT

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHMG-1	MANAGEMENT	4	3	1	70	20	10

Unit – I Management- Objective, Overview, Process, Functions, Skills and Roles, Nature and Significance, Management and Society (Environment), Evolution of Management Thought – Taylor, Fayol, Elton Mayo, FC Bernard, Likert and Webber (Theories of Management), Recent Trends and Future Challenges of Management

Unit – II Understanding Organizational behavior: Levels of analysis within OB – individual, group and organization, challenges and opportunities for OB, relationship of OB with other fields.

Unit-III Concepts and Perspectives on Human Resource Management, Human Resource Management in a changing Environment, Corporate Objective and Human Resource Planning.

Unit – IV Nature and scope of marketing, corporate orientations towards the marketplace. The Marketing Environment and Environment Scanning. Introduction to Consumer Behavior, Consumer Behavior and Marketing Strategy, Consumer Involvement and Decision Making.

Unit – V Management Accounting Concept, Needs, Importance, Cost Accounting –Records and Processes, Inventory Valuation, Reconciliation between Financial and Cost Accounts.

Suggested Readings:

1. Koonz O Donnel: Principles of Management
2. Gupta, Sharma and Bhalla: Principles and Practices of Management, Kalyani Publication
3. Aswathappa, K. 'Human Resource and Personnel Management', Tata McGraw-Hill, New Delhi, 1997.
4. Edwin B. Flippo, 'Personnel Management', New York, Mc Graw Hill Book Company, International Edition, 1984.
5. Srinivasan, R. and Chunawalla, S.A., Management Principles and Practices, Himalaya Publishing House, Mumbai, 1995.
6. Saxena Rajan., 'Marketing Management', Tata McGraw Hill.
7. Sontakki C.N., 'Marketing Management', Kalyani Publication.
8. Bhattacharya S K and Dearden J. 'Accounting for Management'. Text and Cases. New Delhi, Vikas 1996.
9. Sharma & Gupta, 'Management Accounting' Kalyani Publication.
10. Chandan, J., "Organizational Behaviour", Vikas Publishing House Pvt. Ltd.
11. Udai Pareek, "Organizational Behaviour", Oxford University Press, 2004.



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSEWORK

SUBJECT: COMMERCE

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	BBCM-1	COMMERCE	4	3	1	70	20	10

Unit – I Concept of Management, Scope and Nature of Management, Approaches to Management, Human Relation, Behavioral and System approach.

Unit – II Theoretical Framework of Business Environment: Concept, Significance and nature of business environment; Elements of environment -internal and external; Changing dimensions of business environment, Liberalization, Privatization and Globalization.

Unit-III Final Accounts (Advanced), Advanced Problems of Bank Reconciliation Statement, Rectification of Errors and Accounting for Non Profit Organization.

Unit – IV Various cost concepts, Cost centre and cost unit, Methods and techniques of Costing. Installation of costing system, Methods of inventory control, Overheads Accounting.

Unit – V The Companies Act, 1956 (Relevant Provisions): Definition, types of companies, Memorandum of association; Articles of association; Prospectus; Share capital and Membership, Meetings and Resolutions; Company Management; Managerial Remuneration; Winding up and dissolution of companies.

Suggested Readings:

12. Understanding Organization -Shukla
13. Adhikary, M: Economic Environment of Business, Sultan Chand & Sons, New Delhi
14. Ahluwalia, I.J.: Industrial Growth in India, Oxford University Press Delhi.
15. Oswal Maheshwari -Cost Accounting
16. M. Agarwal & M Jain-Cost Accounting
17. Jain& Narang -Cost Accounting
18. Singh, Avtar: Law Relating to Monopolies, Restrictive and Unfair Trade Practices, Eastern Book Co., Lucknow.
19. The Companies Act, 1956



BHABHA UNIVERSITY, BHOPAL

COURSE WORK FOR Ph.D. STUDENT

SUBJECT: ENGLISH

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHEN-1	ENGLISH	4	3	1	70	20	10

Section A 1

Research Methodology

1. What is research
2. Defining the hypothesis
3. Research design
4. Types of research- quantitative and qualitative
5. Identification and evaluation of resources
6. Differences in methods of data collection in quantitative and qualitative type
7. Writing a research proposal ,
 - Preamble/introduction
 - Problem
 - Objectives of the study
 - Hypothesis
 - Study design and structure
 - Problem and limitations
 - Documentation
8. Writing a research report
 - Size and physical design
 - Procedure
 - Data collection
 - Analysis and interpretation/critique of data
 - Findings
 - Conclusion
 - Layout
 - Treatment of quotations
 - Documentation style
9. Role of computer in research work-
 - Identification of area and topic- internet and web resources
 - Collection of data/information for study
 - Drafting and editing of proposal and report
 - Use of text supporting applications like MS Word, Excel, Pagemaker etc.
 - Documentation and referencing

Recommended Reading

1. Adhikari, Madhu Malati. A Students Handbook for Writing Research Term Paper, 2004
2. Gibaldi, Joseph. MLA Handbook for Writers of Research Papers. *Latest edition*
3. Hillways, Tyrus. Introduction to Research. Foster Young Mifflin Co., 1956.
4. Khandelwal, R.L. Research Methodology: A Symposium. Anand Vallabh Vidyapeeth, 1978.
5. Kothari, C. R. Research Methodology: Methods and Techniques. New Delhi: New Age International (P) Ltd., 2010
6. Kumar, Ranjit. Research Methodology. New Delhi: Sage Publications, 2011.
7. Sandros, Chaney. An Introduction to Research in English. New York: The Macmillan Co., 1958.
8. Singh, Inderpal *et al.* Research Methodology and Statistical Methods. Ludhiyana: Kalyani Publishers, 2009.
9. Turabair, K.L. A Manual for writers of Term Papers, Thesis & Dissertation. Chicago: The University of Chicago.



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSEWORK

SUBJECT- SOCIOLOGY

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHSL-1	SOCIOLOGY	4	3	1	70	20	10

1. Philosophical roots of social Research. Enlightenment, Reason & Science, structure of scientific revolution.
2. Logic of enquiry in social sciences research.
3. Scientific method in social research: Observation, interview schedule & questionnaire, content analysis, sampling design, case study method.
4. Survey techniques measurement and scaling.
5. Operationalization & research design.
6. Encounters and experiences in field work.
7. Application of computers in social research (SPSS)

References-

1. Carol Grbich: New Approaches in Social Research, Sage, 2000
2. Williams Malcolm, Science and Social Science, Routledge, New York, 2004.
3. Mulkay Michael: Science and the Sociology of Knowledge, George Allen and Unwin Ltd., London, 1979.
4. Giddens Anthony, New Rules of sociological Research, Hutchinson, 1976.
5. Denzin Norman, Lincoln Yvonna (ed): Handbook of Qualitative Research, Sage, Thousand Oaks, 2000.
6. Silverman David: Qualitative Methodology and sociology, Gower, Vermont, 1985.
7. Burawoy M. and Joseph Blum, (ed): Global Ethnography: Forces, 2000, Connections and Imaginations, University of California Press, Berkeley and Los Angeles.
8. Bryman Alan: Social Research Methods, Oxford, 2001.
9. Devine and Heath: Sociological Research Methods in Context, Palgrave 1999.
10. Sachdev Meetal: Qualitative Research in Social Sciences, Raj Publishing, Jaipur.
11. Feyerabend Paul: Farewell to Reason, Verso, London, 1987.
12. Feyerabend Paul: Against Method, Humanities Press, 1975.
13. Sarantakos S: Social research, Macmillan Press, UK, 1999



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSE WORK SUBJECT- Computer Science Engineering

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHCS-1	Computer Science Engineering	4	3	1	70	20	10

Introduction to Machine Learning

This course provides a concise introduction to the fundamental concepts in machine learning and popular machine learning algorithms. We will cover the standard and most popular supervised learning algorithms including linear regression, logistic regression, decision trees, k-nearest neighbour, an introduction to Bayesian learning and the naïve Bayes algorithm, support vector machines and kernels and neural networks with an introduction to Deep Learning. We will also cover the basic clustering algorithms. Feature reduction methods will also be discussed. We will introduce the basics of computational learning theory. In the course we will discuss various issues related to the application of machine learning algorithms. We will discuss hypothesis space, overfitting, bias and variance, tradeoffs between representational power and learn ability, evaluation strategies and cross-validation. The course will be accompanied by hands-on problem solving with programming in Python and some tutorial sessions.

PREREQUISITES

Basic programming skills (in Python), algorithm design, basics of probability & statistics

Unit 1:

Introduction: Basic definitions, types of learning, hypothesis space and inductive bias, evaluation, cross-validation. Linear regression, Decision trees, overfitting.

Unit 2:

Instance based learning, Feature reduction, Collaborative filtering based recommendation. Probability and Bayes learning.

Unit 3:

Logistic Regression, Support Vector Machine, Kernel function and Kernel SVM. Neural network: Perceptron, multilayer network, backpropagation, introduction to deep neural network.

Unit 4:

Computational learning theory, PAC learning model, Sample complexity, VC Dimension, Ensemble learning.

Unit 5:

Clustering: k-means, adaptive hierarchical clustering, Gaussian mixture model.

References:

1. Machine Learning. Tom Mitchell. First Edition, McGraw- Hill, 1997.
2. Introduction to Machine Learning Edition 2, by Ethem Alpaydin



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSE WORK SUBJECT- Electronics and Communication Engineering

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHEC-1	Electronics and Communication Engineering	4	3	1	70	20	10

UNIT-I: Analog Circuits

Small Signal Equivalent circuits of diodes, BJTs, MOSFETs and analog CMOS. Simple diode circuits, clipping, clamping, rectifier. Biasing and bias stability of transistor and FET amplifiers. Amplifiers: single- and multi-stage, differential and operational, feedback, and power. Frequency response of amplifiers. Simple op-amp circuits. Filters. Sinusoidal oscillators; criterion for oscillation; single- transistor and op-amp configurations. Function generators and wave-shaping circuits, 555 Timers. Power supplies.

UNIT-II: Digital circuits

Boolean algebra, minimization of Boolean functions; logic gates; digital IC families (DTL, TTL, ECL, MOS, CMOS). Combinatorial circuits: arithmetic circuits, code converters, multiplexers, decoders, PROMs and PLAs. Sequential circuits: latches and flip-flops, counters and shift-registers. Sample and hold circuits, ADCs, DACs.

UNIT-III: Networks

Network graphs: matrices associated with graphs; incidence, fundamental cut set and fundamental circuit matrices. Solution methods: nodal and mesh analysis. Network theorems: superposition, Thevenin and Norton's maximum power transfer, Wye-Delta transformation. Linear constant coefficient differential equations; time domain analysis of simple RLC circuits, Solution of network equations using Laplace transform: frequency domain analysis of RLC circuits. 2-port network parameters: driving point and transfer functions.

UNIT-IV: Control Systems

Basic control system components; block diagrammatic description, reduction of block diagrams. Open loop and closed loop (feedback) systems and stability analysis of these systems. transient and steady state analysis of LTI control systems and frequency response. Tools and techniques for LTI control system analysis: root loci, Routh-Hurwitz criterion, Bode plots.

UNIT-V: Communications

Random signals and noise: probability, random variables, probability density function, autocorrelation, power spectral density. Analog communication systems: amplitude and angle modulation and demodulation systems, spectral analysis of these operations, superheterodyne receivers; elements of hardware, realizations of analog communication systems; signal-to-noise ratio (SNR) calculations for amplitude modulation (AM) and frequency modulation (FM) for low noise conditions.

Reference Books:

- 1- Jacob Millman and Christos Halkias, "Integrated Electronics" Tata McGraw Hill.
- 2- R.P. Jain, "Modern Digital Electronics", Tata McGraw Hill
- 3- D Roy Chaudhary, "Network and System", New Age International
- 4- I. J. Nagrath and M. Gopal, "Control System Engineering", New Age International
- 5- George Kannedy and Bernard Davis, "Electronics Communication System", Tata McGraw Hill



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSE WORK

SUBJECT- PHARMACY

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHPY-1	PHARMACY	4	3	1	70	20	10

1. Preformulation studies: Preformulation studies of drug substances, proteins and peptides. Preformulation work sheet.
2. Complexation: Metal and organic molecular complexes, inclusion compounds with reference to cyclodextrins, methods of analysis.
3. Solubilization: Solubility and solubilization of nonelectrolyte, drug solubilization in surfactant systems, use of co-solvents, solid-state manipulations and drug derivitization.
4. Optimization: Statistical methods and factorial design, Quality By Design.
5. Stability : Stability of dosage forms as per ICH guidelines
6. Solid State Pharmaceutics Molecular Level : Crystallinity , crystal habit, polymorphism, amorphous state, solvates, hydrates, analytical techniques for characterization(DSC, PXRD, SEM, FTIR), molecular modeling in solid state characterization- case studies and regulatory perspective Particle level : Particle size, particle shape, porosity, surface area, compaction, particle engineering in pharmaceuticals and relevance in doses form designing Bulk level : Bulk density, compressibility, flow properties, compaction and consolidation cohesivity, electrostatics, aggregation, agglomeration, role in formulation development and processing



BHABHA UNIVERSITY, BHOPAL,

SYLLABUS OF Ph.D. COURSE WORK SUBJECT- EDUCATION

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHED-1	EDUCATION	4	3	1	70	20	10

Unit 1 Philosophical foundation of Education

Indian
Schools
Western
schools

Unit 2 Sociological foundation of Education

Relationship of Sociology and Education
Sociological Aspects of Education

Unit 3 Psychological foundation of Education

Motivation and learning theories.
Intelligence- types, theories and measurement
Personality- types, theories and measurement
Inclusive Education

Unit 4 Educational Evaluation and Assessment

Tools of Evaluation
Characteristics of good measuring instrument
Test standardization
Continuous and comprehensive evaluation

Unit 5 Educational Technology

Multimedia Approach in teaching & learning
Instructional system design
E-communication

Unit 6 Teacher Education

Innovations in TE
Quality Management in TE

Unit 7 Data Analysis

Qualitative and Quantitative data Analysis
Descriptive analysis and Inferential Analysis



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSE

WORK SUBJECT- ELECTRICAL ENGINEERING

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHEE-1	ELECTRICAL ENGINEERING	4	3	1	70	20	10

Unit 1

Protective Relays: Relaying review, characteristics and operating equations of relays. CT's and PT's differential relay, over-current relay, reverse power relay, distance relays, applications of relays.

Unit 2

STATIC RELAYS: Introduction, advantages and disadvantages, classification logic ckts, smoothing circuits, voltage regulator square wave generator, time delay ckts level detectors, summation device, sampling circuit, zero crossing detector, output devices. COMPARATORS: Replica Impedance, mixing transformers, general equation of phase and amplitude comparator, realization of ohm, impedance and off set impedance characteristics, duality principle, static amplitude comparators, coincidence circuit, Hall effect devices, Magneto receptivity, zener diode phase comparator multi input comparators.

Unit 3

Generator and transformer protection: Protective devices for system. Protective devices for stator, rotor, and prime mover of generator, percentage differential relays protection, three winding transformer protection, earth fault protection, generator transformer unit protection.

UNIT 4

Bus bar and transmission line protection: Distance protective schemes, directional wave detection relay. Phase compensation carrier protection. High impedance differential scheme, supervisory and check relay, Some features of 500 KV relaying protection.

Unit 5

Modern trends in power system protection: Different types of digital and computer aided relays, Microprocessor based relays, auto-reclosing, frequency relays, under and over frequency relays, di/dt relays. Algorithms for transmission line, transformer & bus bar protection; out-of-step relaying Introduction to adaptive relaying & wide area measurements

Reference Books:

1. Power System Protection and Switchgear, B.Ram – Tata Mc-Graw Hill Pub.
2. Switchgear and Protection, M.V.Deshpande - Tata Mc-Graw Hill Pub.
3. Power System Protection & Switchgear, Ravindra Nath, M.Chander, Willy P
4. Computer Relaying for power system, Arun Phadke, James Thorp, Johns W P
5. Power System Protection, M.A.Date, Bharti Prakashan, Vallabh Vidya N,(Guj).



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSE WORK SUBJECT- HINDI

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHHN-1	HINDI	4	3	1	70	20	10

इकाई – 1 साहित्य का स्वरूप : परम्परा एवं आधुनिक संदर्भ, साहित्य का प्रयोजन, साहित्य की प्रासंगिकता, विचारधारा और साहित्य।

इकाई – 2 हिन्दी साहित्य का आदिकालीन काव्य (नाथ, सिद्ध, जैन, रासो एवं लौकिक साहित्य) मध्यकालीनबोध का स्वरूप, भक्तिकाल की सामाजिक और सांस्कृतिक परिस्थितियाँ, भक्ति आंदोलन और लोकचेतना, विभिन्न दार्शनिक मत—अद्वैतवाद, विशिष्टाद्वैत, द्वैतवाद, द्वैताद्वैतवाद, शुद्धाद्वैतवाद।

इकाई – 3 सूफीमत का विकास, सूफी काव्य में भारतीय संस्कृति और लोक जीवन, कबीर और जायसी का साहित्यिक एवं सांस्कृतिक अवदान, संत कवियों की लोकचेतना।

इकाई – 4 रामभक्ति काव्य और कृष्ण भक्ति काव्य में चित्रित भारतीय संस्कृति और लोकजीवन, सूरदास और लोकदृष्टि, पुष्टिमार्ग के सिद्धांत, तुलसीदास का समन्वयवाद, सगुण भक्तिकाव्य का साहित्यिक और सांस्कृतिक अवदान, रीतिकालीन कवियों की लोकचेतना।

इकाई – 5 मार्क्सवाद, मनोविश्लेषणवाद, अस्तित्ववाद, संरचनावाद, विखण्डनवाद, आधुनिकता, उत्तर आधुनिकता, समाजवाद।



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSEWORK

SUBJECT- HISTORY

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHHT-1	HISTORY	4	3	1	70	20	10

1. अनुसंधान की प्रकृति – विशुद्ध एवं व्यवहारिक अनुसंधान
Nature of Research, Pure and Applied Research
2. वैज्ञानिक पद्धति – सामान्य शोध और विज्ञान, वैज्ञानिक चिंतन के चरण –
वैज्ञानिक पद्धति और इतिहास
The Scientific Method - Common Sense and Science Steps in Scientific Thinking. The Scientific Method and History.
3. अध्ययन की प्रकृति : व्यक्ति सूची अध्ययन, वैयक्तिक अध्ययन, क्षेत्रीय अध्ययन।
Nature of Study : Pannel Studies, Case Studies and Area Studies.
4. अनुसंधान प्ररचना : उपकल्पनायें और अवधारणायें
Research Fomulation : Hypothesis and Concepts.
5. तथ्य सामग्री के स्रोतों के प्रकार – प्राथमिक और द्वितीयक तथ्य सामग्री के
विशेष संदर्भ में
Types of Sources of Data with Special Reference to Primary and Secondary Data.
6. तथ्य सामग्री संकलन की प्रविधियां : सामग्री विश्लेषण – अवलोकन,
प्रश्नावलियां और अनुसूचियां, यांत्रिक साधनों का प्रयोग – टेपरिकार्डर, पंचर,
बेरीफायर्स, सोर्टर आदि.
The Techniques of Data Collection : Content Analysis, Observation - Questionnaires and Schedules use of Mechanical Aldus - Tap Recorder, Puncher, Verifiers, Sorter etc.
7. क्षेत्रीय कार्य की प्रविधियां – सहभागी और असहभागी निरीक्षण, साक्षात्कार,
जीवन इतिहास का प्रयोग आदि.
Techniques of Field Work - Participant and Non Participant Observation, Interviewing, use of Life Histories etc.
8. तथ्य विश्लेषण और प्रतिवेदन-लेख
Data Analysis and Report - Writing



BHABHA UNIVERSITY, BHOPAL

SYLLABUS OF Ph.D. COURSE WORK

SUBJECT- PHYSICS

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHPH-1	PHYSICS	4	3	1	70	20	10

Unit-I: General: Definition of Research, Components of Research Problems, Scientific Hypotheses, Research Purpose, Research Design, Literature searching, Aims and Objectives, Expected outcome, Methodology to be adapted, Planning of experiments/theory.

Unit-II: Data Collection & Analysis: Types of data methods and techniques of data collection, primary and secondary data, meta-analysis, historical methods, content analysis, devices used in data collection, pilot study and protest of tools, choice of data collection methods.

Unit-III: Basics of Computers: Operating System-Windows and Linux, MS OFFICE (Document, EXCEL, power point Presentation), Peripheral device installation and operation, antivirus, internet browsing.

Unit-IV: Paper writing and Report Generation: Basic concept of paper/ thesis writing and report generation, writing Research abstract, Introduction, review of literature, Result, conclusion, concepts of Bibliography and References, significance of report writing steps of report writing, types of research reports, method of presentation of report, formats of publication in research general. **References:**

1. "How to write & Publish" by Robert A.Day and Barbara Gastel, (Cambridge University Press)
2. "Survival skills for Scientists" by Federico Rosel & Tudor Johnson (Imperial College Press).
3. "How to Research" by Loraine Blaster, Christina Hughes & Malcum Tight (Viva Books)
4. "Probability and Statistics for Engineers and Scientists" by Sheldon Ross (Elsevier Academic Press).
5. "The Craft of Scientific Writing" by Michael Alley (Springer)
6. "A Students Guide to Methodology" by Peter Clough and Cathy Nutbrown (Sage Publications.)
7. Kothari, C.R., Research Methodology (Methods & Techniques), New Age Publishers.



BHABHA UNIVERSITY, BHOPAL

Ph. D. course work

Subject: LIBRARY INFORMATION SCIENCE

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHLS-1	LIBRARY INFORMATION SCIENCE	4	3	1	70	20	10

1. Research: General Meaning, need, process Spiral scientific method

Kinds of Research: Diagnostic, Descriptive, Exploratory, Explanatory Research Ethics

2. Typology for Literature search.

Typology for Literature search

Scientific methods: Components of scientific methods. Formulation of research problems

3. Research Design

Types of research design: Historical design, Descriptive design, Formation of Hypothesis; Synopsis Writing

4. Brief Review of Research Methods

5. Qualitative & Quantitative Techniques

6. Analysis, Interpretation & Presentation of data

Presentation of Data Statistical analysis

Use of Statistical package Use of

Computers in research

7. Research Reporting

Style manuals (Chicago, APA, MLA) Evaluation of research

Current trends in LIS research



BHABHA UNIVERSITY, BHOPAL

Pre Ph.D. Syllabus

Subject: Zoology

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHZO-1	Zoology	4	3	1	70	20	10

Unit-1- Physiology & Animal behavior- Thermoregulation, Digestion and Respiration. Circulation, Excretion and Reproduction. Nervous system- CNS, PNS, ANS and Sense organs. Neuroendocrine system- Endocrine organs, classification of hormones and mechanism of hormone action. Stress and adaptations. Homeostasis. Feeding learning, social and sexual behavior. Circadian rhythms, parental care and mimicry. Migration of insects, fishes and birds. Methods in field biology – methods in study of behavior, habit characterization, ground and remote sensing methods.

Unit-2- Evolution & Developmental biology- Principles and methods of taxonomy, levels of structural organization Concept and theories of evolution. Organic evolution. Origin of life and origin of species. Cell differentiation and differential gene activity, Totipotency, Organizers- Concepts and induction process. Biomarkers of cancer detection.

Unit-3- Cellular and molecular biology- Cell cycle, Stem cells and regenerative medicine, Gene and gene action, DNA damage and repair.

Unit-4- Advances and Recent tools in Biology- Electrophoresis and chromatographic techniques for characterization of Biomolecules DNA, RNA, Protein and Enzymes. DNA Microarray, DNA sequencing and Molecular phylogeny, Methods of gene transfer, GISH (Genomic In Situ Hybridization) and FISH (Fluorescence In Situ Hybridization). Flow cytometry, Polymerase Chain Reaction (PCR), ELISA, Northern, Western, Southern Blot techniques. DNA barcoding and systemic classification, RFLP, RIA, RAPD and AFLP. 8) Immunofluorescence

Unit-5- Computational methods & Environmental Science- Proteomics and genomic analysis, nucleic acid and protein sequencing Databases, Data mining for sequence analysis, Web based tools for sequence search. Biodiversity- Types and conservation strategies. Ecosystem and types. Biotic, abiotic factors and their interrelation. Ecological succession and types. Natural resources, types and their conservation. Wild life management and conservation acts. Global warming, Bioremediation, Biosensor, Bioresources.

Reference books:

- 1- C.B. Pawar: Genetics Vol I and II, Himalaya Publishing House, Mumbai
- 2- Crop pests and how to fight them- Govt. of Maharashtra Pub. Bombay
- 3- Dr. Nageshwar Rao and Dr. Rajendra P. Das: Communication Skills, Himalaya Publishing House 2005
- 4- Dr. P.V. Jabade: General Physiology
- 5- G. J. Tortora: Principle of Anatomy and Physiology
- 6- B. Lewin: Genes Xth edition, Wiley Eastern Limited, New Delhi
- 7- B.K. Berry: Animal Physiology
- 8- C.C. Chatterjee: Human Physiology
- 9- Dubey, R. C. - A text book of Biotechnology, S. Chand and Company Ltd, New Delhi, 1996.
- 10- Bell and Davidson: Textbook of Physiology and Biochemistry.



BHABHA UNIVERSITY, BHOPAL

Pre Ph.D. Syllabus

Subject: Research Methodology

Sr. No.	Subject Code	Subject Name	Credit	L	T	End Semester Exam	Tests (Two)	Assignments/Quiz/Attendance
1	PHRM-1	Research Methodology	4	3	1	70	20	10

Unit – I Introduction to Research Methodology - Meaning, Objectives, Significance of Research in Management, Importance and scope of Research Methodology.

Unit – II Research Process - Defining Research Problem, Setting of Hypothesis, Research Design – Exploratory, Descriptive and Experimental Research Designs.

Unit – III Sampling Design - Steps in Sampling Design, Criteria of Selecting a Sampling Procedure, Characteristics of Good Sample Design, and Types of Sample Design.

Unit – IV Data Collection-Primary and Secondary data, Observational and Survey Methods, Questionnaire Design, Processing of Data – Editing, Coding, Classification, Field Work and Tabulation of Data.

Unit – V Analysis and Report Writing - Selection of Appropriate Statistical Techniques, Parametric Test for hypothesis testing – t - test, Chi – Square test, Characteristics of Non Parametric Test, One-way ANOVA, Report writing.

Suggested Readings:

1. Bennel, Roger, 'Management Research', ILO,
2. Gupta, S.P. 'Statistical Methods', 30th ed., Sultan Chand, New Delhi, 2001.
3. Golden, Biddle, Koren and Karen D. Locke, 'Composing Qualitative Research', Sage Pub. 1997.
4. Kothari C.R., 'Research Methodology', Wishwa Prakashan.